ILT Research System - System Architecture & Technical Specifications (v5.3)

Infrastructure Architecture, Analyst Replay Engine, Open-Source BI Integration, Unlimited Templates, and Full DDL Schema

Specification Parameter	Design Specification	Specification Parameter	Design Specification
Document ID:	ILT-ARCH-SPEC-V5.3-EN	Release Date:	October 2026 (Release 5.3 Full Edition)
Clinical Basis:	Prof. Adi-Japha, Dr. Hollander, Dr. Dorfberger	Hardware:	Wacom MovinkPad 11 (Certified Continuous Charging)
Analyst Infrastructure:	Replay Engine + Apache Superset + DuckDB	Video Subsystem:	Camera-Agnostic Ingestion (RTSP/ONVIF/USB)
Database:	PostgreSQL 16 + TimescaleDB + MinIO	Clinical Metrics:	3 Distinct Metrics + Timer Factoring

1. Executive Architectural Summary (Release 5.3)

- **Analyst Portal & Kinematic Replay Architecture:** Dedicated API layer streaming EMR time-series from TimescaleDB to Canvas Replayer, with time-code synchronization against video files in MinIO.
- **Open-Source BI & Analytical Engine Integration:** View v_analyst_cohort_slices, native connection pooling for Apache Superset and Metabase, embedded JupyterHub server, and high-speed DuckDB engine exporting Parquet.
- **Agnostic Video Ingestion & Unlimited Templates Engine:** Agnostic RTSP/ONVIF/USB ingestion; table dynamic_templates with JSON schemas for all elementary school rulings.

2. Relational Database Schema & Analytical Views (DDL v5.3)

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-- ================================================================================
-- ILT PRODUCTION DDL (PostgreSQL 16 + TimescaleDB) - Release 5.3 / DDD v1.2
-- ================================================================================
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";
CREATE EXTENSION IF NOT EXISTS "timescaledb";

-- 1. System Configuration & Global Defaults
CREATE TABLE system_config (
    config_id INT PRIMARY KEY DEFAULT 1,
    in_transition_delay_sec INT NOT NULL DEFAULT 20, -- Default In-Transition: 20s
    inactivity_timeout_sec INT NOT NULL DEFAULT 20,  -- Default Inactivity Timeout: 20s
    post_block_timeout_sec INT NOT NULL DEFAULT 30,  -- Timeout post-15th letter: 30s
    notebook_page_timeout_sec INT NOT NULL DEFAULT 60,
    video_recording_enabled BOOLEAN NOT NULL DEFAULT TRUE,
    require_research_director_approval BOOLEAN NOT NULL DEFAULT TRUE,
    score_weight_pen_lifts_pct NUMERIC(5,2) NOT NULL DEFAULT 35.00,
    score_weight_accuracy_pct NUMERIC(5,2) NOT NULL DEFAULT 40.00,
    score_weight_time_pct NUMERIC(5,2) NOT NULL DEFAULT 25.00,
    CONSTRAINT chk_weights_sum_100 CHECK (
        (score_weight_pen_lifts_pct + score_weight_accuracy_pct + score_weight_time_pct) = 100.00
```

```
    ),
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 2. Domains Table (with Research Director Authorization for Overrides)
CREATE TABLE domains (
    domain_id SERIAL PRIMARY KEY,
    domain_name VARCHAR(100) NOT NULL UNIQUE,
    override_in_transition_sec INT NULL,
    override_inactivity_timeout_sec INT NULL,
    override_video_enabled BOOLEAN NULL,
    timer_approved_by_research_director VARCHAR(100) NULL,
    timer_approval_timestamp TIMESTAMP WITH TIME ZONE NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 3. Dynamic Templates Engine
CREATE TABLE dynamic_templates (
    template_id VARCHAR(50) PRIMARY KEY,
    template_name VARCHAR(100) NOT NULL,
    template_category VARCHAR(30) NOT NULL,
    ruling_type VARCHAR(30) NOT NULL,
    target_grade_level VARCHAR(30) DEFAULT 'ALL_GRADES',
    canvas_schema JSONB NOT NULL,
    rules_schema JSONB NOT NULL,
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 4. Canonical Models for Freehand Row 3 Evaluation
CREATE TABLE letter_canonical_models (
    letter_id VARCHAR(10) PRIMARY KEY,
    language_code VARCHAR(5) NOT NULL,
    letter_char VARCHAR(5) NOT NULL,
    expected_strokes_count INT NOT NULL DEFAULT 1,
    canonical_aspect_ratio NUMERIC(5,3) NOT NULL,
    canonical_points JSONB NOT NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 5. Sessions Table (3 Metrics Preserved Independently + Timer Snapshots)
CREATE TABLE sessions (
    session_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    session_number BIGSERIAL UNIQUE,
    domain_id INT NOT NULL REFERENCES domains(domain_id),
    school_name VARCHAR(150) NOT NULL,
    grade_class VARCHAR(50) NOT NULL,
    evaluator_name VARCHAR(120) NOT NULL,
    assessment_date DATE NOT NULL DEFAULT CURRENT_DATE,
    examinee_alias VARCHAR(50) NOT NULL,
    examinee_initials VARCHAR(5) NOT NULL,
    student_language VARCHAR(5) DEFAULT 'he',
    session_stage VARCHAR(30) NOT NULL,
    is_charging_during_test BOOLEAN DEFAULT FALSE,
```

```
    sys_in_transition_sec INT NOT NULL DEFAULT 20,
    dom_in_transition_sec INT NOT NULL DEFAULT 20,
    sys_inactivity_sec INT NOT NULL DEFAULT 20,
    dom_inactivity_sec INT NOT NULL DEFAULT 20,
    video_recording_enabled BOOLEAN NOT NULL DEFAULT TRUE,
    warmup_row_passed BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 6. Assessment Blocks Table (Preserving 3 Distinct Metrics Independently)
CREATE TABLE session_blocks (
    block_id BIGSERIAL PRIMARY KEY,
    session_id UUID NOT NULL REFERENCES sessions(session_id) ON DELETE CASCADE,
    block_type VARCHAR(30) NOT NULL,
    block_index INT NOT NULL,
    target_letter VARCHAR(5) NULL,
    pen_lifts_count INT NOT NULL,
    accuracy_pct NUMERIC(5,2) NOT NULL,
    total_duration_ms INT NOT NULL,
    composite_score NUMERIC(6,2) NULL,
    is_repeated_disrupted BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 7. High-Frequency EMR Telemetry (TimescaleDB Hypertable with 1-Day Chunks)
CREATE TABLE pen_telemetry_chunks (
    recorded_at TIMESTAMP WITH TIME ZONE NOT NULL,
    session_id UUID NOT NULL,
    block_index INT NOT NULL,
    house_id INT NOT NULL,
    time_offset_ms INT NOT NULL,
    x NUMERIC(8,4) NOT NULL,
    y NUMERIC(8,4) NOT NULL,
    pressure NUMERIC(6,5) NOT NULL,
    tilt_altitude SMALLINT NOT NULL,
    tilt_azimuth SMALLINT NOT NULL,
    is_hover BOOLEAN NOT NULL,
    is_pen_down BOOLEAN NOT NULL
);
SELECT create_hypertable('pen_telemetry_chunks', 'recorded_at', chunk_time_interval => INTERVAL '1
day');
ALTER TABLE pen_telemetry_chunks SET (timescaledb.compress, timescaledb.compress_segmentby =
'session_id, block_index');
SELECT add_compression_policy('pen_telemetry_chunks', INTERVAL '7 days');

-- 8. Video Replay Records (Desk & Hands RTSP / MP4 Ingestion)
CREATE TABLE session_video_records (
    video_id BIGSERIAL PRIMARY KEY,
    session_id UUID NOT NULL REFERENCES sessions(session_id) ON DELETE CASCADE,
    camera_model VARCHAR(100) DEFAULT 'Wireless IP Camera (RTSP)',
    storage_bucket VARCHAR(50) DEFAULT 'ilt-session-videos',
    video_object_key TEXT NOT NULL,
    duration_sec NUMERIC(8,2) NOT NULL,
    start_time_utc TIMESTAMP WITH TIME ZONE NOT NULL,
```

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    fps NUMERIC(4,2) DEFAULT 25.00,
    resolution VARCHAR(20) DEFAULT '1280x720',
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 9. Analytical Cohort Slices View (for Apache Superset, Metabase, DuckDB, Jupyter)
CREATE OR REPLACE VIEW v_analyst_cohort_slices AS
SELECT
    s.session_id,
    s.session_number,
    s.examinee_alias,
    s.examinee_initials,
    d.domain_name,
    s.school_name,
    s.grade_class,
    s.dominant_hand,
    s.student_language,
    s.session_stage,
    s.assessment_date,
    s.sys_in_transition_sec,
    s.dom_in_transition_sec,
    (s.dom_in_transition_sec::numeric / s.sys_in_transition_sec::numeric) AS timer_factoring_ratio,
    sb.block_id,
    sb.block_type,
    sb.block_index,
    sb.target_letter,
    sb.pen_lifts_count,
    sb.accuracy_pct,
    sb.total_duration_ms,
    (sb.total_duration_ms::numeric / 1000.0) AS total_duration_sec,
    ((sb.total_duration_ms::numeric / 1000.0) * (s.sys_in_transition_sec::numeric /
s.dom_in_transition_sec::numeric)) AS factored_duration_sec,
    sb.composite_score,
    sb.is_repeated_disrupted
FROM sessions s
JOIN domains d ON s.domain_id = d.domain_id
JOIN session_blocks sb ON s.session_id = sb.session_id;
```

3. Architectural Subsystems & Analyst Workstation

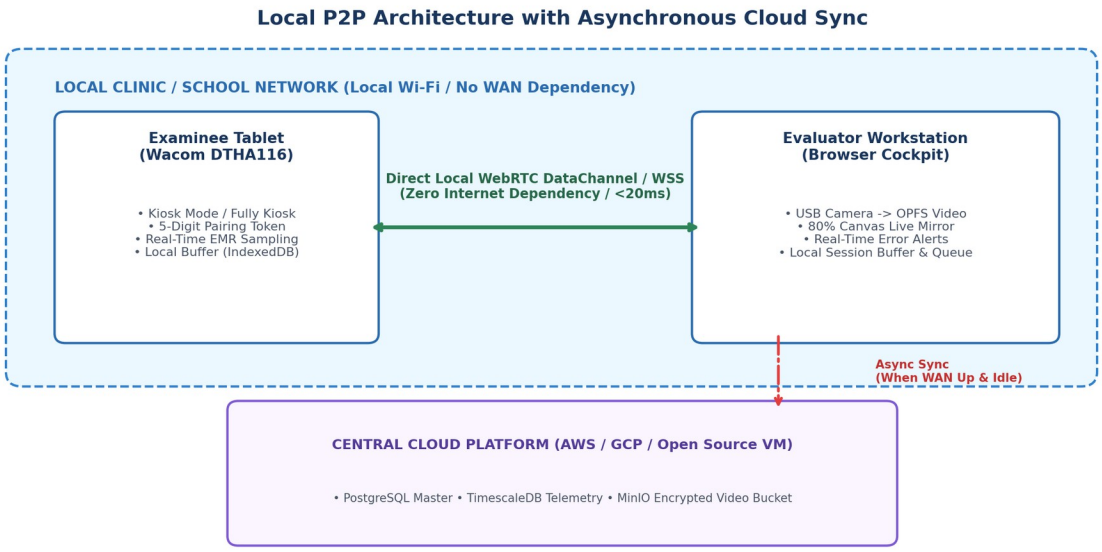


Figure 1: Local P2P Network Topology & Sync

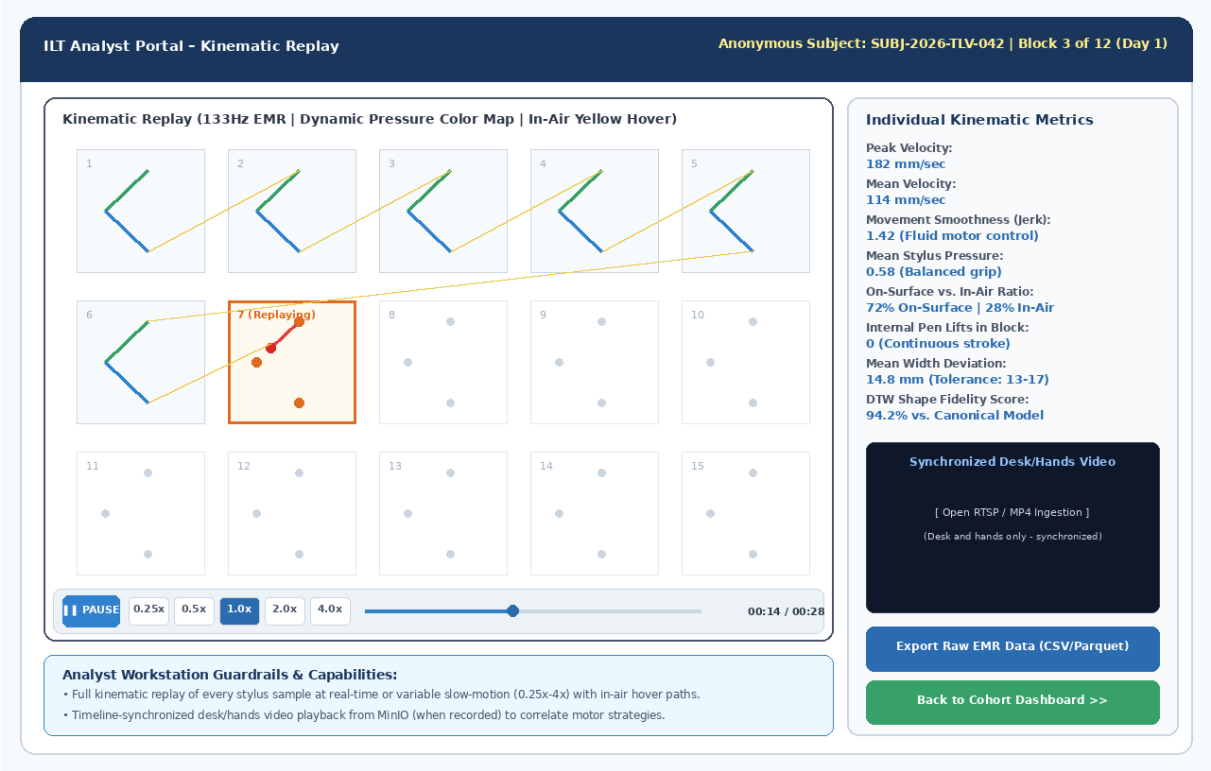


Figure 2: Kinematic Session Replay Engine Architecture Synced with Desk Video

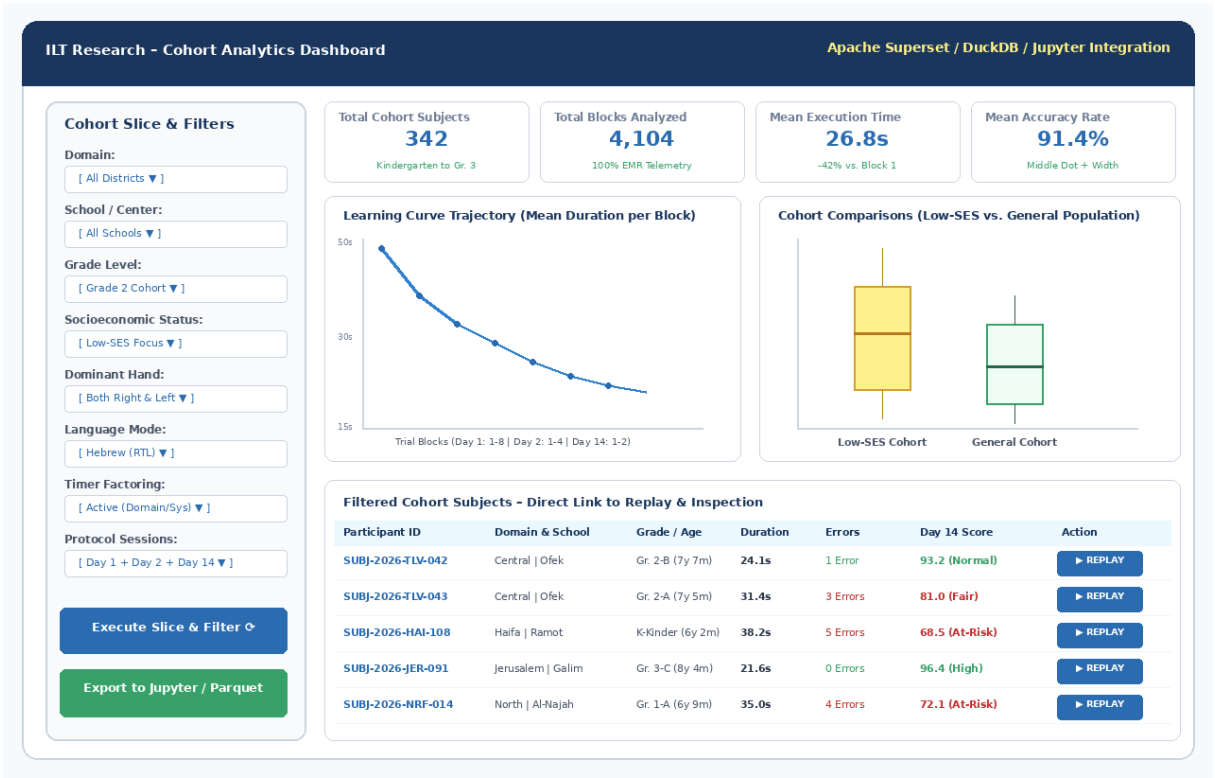


Figure 3: Open-Source Cohort Slice-and-Dice Analytics Dashboard